Aufgabe 01

- $f(x) = 4x^3 \rightarrow f'(x) = 12x^2$
- $f(x) = 4x^3 + x^2 + x + 1 \rightarrow f'(x) = 12x^2 + 2x + 1$
- $f(x) = \sqrt{x} \rightarrow f'(x) = \frac{1}{2\sqrt{x}}$
- $f(x) = \sqrt[4]{x^3} = (x^3)^{1/4} = x^{3/4} \rightarrow f'(x) = \frac{3}{4} x^{-1/4} = \frac{3}{4\sqrt[4]{x}}$
- $f(t) = t^2 e^{3t} \rightarrow f'(t) = 2t e^{3t} + 3t^2 e^{3t} = (2t + 3t^2) e^{3t}$
- $f(x) = (x^n + 1)\sqrt{x^2 - 1} \rightarrow f'(x) = (n x^{n-1} \;)\sqrt{x^2-1} + (x^n+1) \times \frac{2x}{2\sqrt{x^2-1}}$
  $= (n x^{n-1} \;)\sqrt{x^2-1} + \frac{(x^n+1)}{\sqrt{x^2-1}} \cdot x$
- $f(x) = \frac{10x}{x^2+1} \rightarrow f'(x) = \frac{10(x^2+1) - 20x^2}{(x^2+1)^2} = \frac{-10x^2 + 10}{(x^2+1)^2} = \frac{10(1-x^2)}{(x^2+1)^2}$
- $f(x) = \frac{1+x}{1+x^2} \rightarrow f'(x) = \frac{1 \cdot (1+x^2) - 2x(1+x)}{(1+x^2)^2} = \frac{1 - 2x - x^2}{(1+x^2)^2}$
- $f(x) = 5(x^3 - x^2 + 1)^5 \rightarrow f'(x) = 5 \cdot 5 (3x^2 - 2x)(x^3 - x^2 + 1)^4$
  $= 25 (3x^2 - 2x)(x^3 - x^2 + 1)^4$
- $f(x) = \left(\frac{1+x}{x}\right)^n \rightarrow f'(x) = n\left(\frac{x - (1+x)}{x^2}\right)\left(\frac{1+x}{x}\right)^{n-1} = -\frac{n}{x^2}\left(\frac{1+x}{x}\right)^{n-1}$
- $f(x) = 2x\sqrt{x^2 - 1} \rightarrow f'(x) = 2\sqrt{x^2-1} + 2x \cdot \frac{2x}{2\sqrt{x^2-1}} = 2\sqrt{x^2-1} + \frac{2x^2}{\sqrt{x^2-1}}$
- $u(t) = u_0 \sin(\omega t + \varphi) \rightarrow u'(t) = \omega \cdot u_0 \cos(\omega t + \varphi)$
- $f(x) = \tan(x) = \frac{\sin(x)}{\cos(x)} \rightarrow f'(x) = \frac{\cos^2(x) + \sin^2(x)}{\cos^2(x)} = \frac{1}{\cos^2(x)}$
- $f(x) = a^x = e^{x \ln(a)} \rightarrow f'(x) = (x \ln(a))' \times e^{x \ln(a)} = \ln(a) \cdot (e^{\ln(a)})^x = \ln(a) \cdot a^x$
- $f(x) = \sqrt{a^2 - x^2} \quad f'(x) = \frac{-2x}{2\sqrt{a^2-x^2}} = \frac{-x}{\sqrt{a^2-x^2}}$
- $f(x) = \sin(2x) \rightarrow f'(x) = 2\cos(2x)$
- $f(x) = \frac{a}{x}\sqrt{a^2 + x^2} \rightarrow f'(x) = \frac{-a}{x^2}\sqrt{a^2+x^2} + \frac{a}{x} \cdot \frac{2x}{2\sqrt{a^2+x^2}} = \frac{-a}{x^2}\sqrt{a^2+x^2} + \frac{a}{\sqrt{a^2+x^2}}$
  $= a\left(\frac{1}{\sqrt{a^2+x^2}} - \frac{\sqrt{a^2+x^2}}{x^2}\right) = a\left(\frac{x^2 - (a^2+x^2)}{x^2\sqrt{a^2+x^2}}\right)$
  $= \frac{-a^3}{x^2\sqrt{a^2+x^2}}$

- $f(x) = \frac{1}{\sin(x)} \rightarrow f'(x) = -\frac{\cos(x)}{\sin^2(x)}$
- $f(x) = \cos^3(x) \rightarrow f'(x) = 3(-\sin(x)).\cos^2(x) = -3\sin(x)\cos^2(x)$
- $f(x) = \sin(x^3) \rightarrow f'(x) = 3x^2\cos(x^3)$
- $f(x) = \tan^3(x) + 3\tan(x) \rightarrow f'(x) = 3.\frac{1}{\cos^2(x)} \times \tan^2(x) + 3.\frac{1}{\cos^2(x)}$

$$= \frac{3}{\cos^2(x)}.\left(1 + \tan^2(x)\right) = \frac{3}{\cos^2(x)}.\left(1 + \frac{\sin^2(x)}{\cos^2(x)}\right)$$

$$= \frac{3}{\cos^2(x)}\left(\frac{\cos^2(x)+\sin^2(x)}{\cos^2(x)}\right)$$

$$= \frac{3}{\cos^4(x)}$$

- $f(x) = \tan\left(\frac{1}{x}\right) \rightarrow f'(x) = -\frac{1}{x^2} \times \frac{1}{\cos^2\left(\frac{1}{x}\right)} = \frac{-1}{(x\cos(1/x))^2}$
- $f(x) = \frac{x^2 - a^2}{\sin(x)} \rightarrow f'(x) = \frac{2x.\sin(x) - \cos(x)(x^2-a^2)}{\sin^2(x)}$
- $f(x) = \tan(x^2).x^2 \rightarrow f'(x) = 2x \times \frac{1}{\cos^2(x^2)} \times x^2 + \tan(x^2).2x$

$$= \frac{2x^3}{\cos^2(x^2)} + \tan(x^2)2x.$$

- $f(x) = 3x + \sin(\cos(x)) \rightarrow f'(x) = 3 + (-\sin(x)) \times \cos(\cos(x))$

$$= 3 - \sin(x)\cos(\cos(x))$$

- $f(x) = \sqrt[5]{x^4 - x^2} = (x^4 - x^2)^{1/5} \rightarrow f'(x) = \frac{1}{5}(4x^3 - 2x)(x^4 - x^2)^{-4/5}$
- $f(x) = \sqrt[5]{(x-6)^4} = ((x-6)^4)^{1/5} = (x-6)^{4/5} \rightarrow f'(x) = \frac{4}{5} \times (1).(x-6)^{-1/5}$

$$= \frac{4}{5(x-6)^{1/5}} = \frac{4}{5\sqrt[5]{x-6}}$$

- $f(x) = e^{-x}x^2 \rightarrow f'(x) = -1.e^{-x}.x^2 + e^{-x}.2x$

$$= e^{-x}(-x^2 + 2x)$$

Aufgabe 02

(a) $f(x) = ax + b$

$$f'(x) = \lim_{\Delta x \to 0} \frac{f(x+\Delta x) - f(x)}{\Delta x}$$

$$= \lim_{\Delta x \to 0} \frac{a(x+\Delta x) + b - (ax+b)}{\Delta x}$$

$$= \lim_{\Delta x \to 0} \frac{a \cdot \Delta x}{\Delta x} = \lim_{\Delta x \to 0} a$$

$$= a$$

(b) $f(x) = \sqrt{x}$

$$f'(x) = \lim_{\Delta x \to 0} \frac{(\sqrt{x+\Delta x} - \sqrt{x})}{\Delta x} \cdot \frac{(\sqrt{x+\Delta x} + \sqrt{x})}{(\sqrt{x+\Delta x} + \sqrt{x})}$$

$$= \lim_{\Delta x \to 0} \frac{(\sqrt{x+\Delta x})^2 - (\sqrt{x})^2}{\Delta x (\sqrt{x+\Delta x} + \sqrt{x})}$$

$$= \lim_{\Delta x \to 0} \frac{x + \Delta x - x}{\Delta x (\sqrt{x+\Delta x} + \sqrt{x})}$$

$$= \lim_{\Delta x \to 0} \frac{1}{\sqrt{x+\Delta x} + \sqrt{x}}$$

$$= \lim_{\Delta x \to 0} \frac{1}{2\sqrt{x}}$$

Aufgabe 03

$f(x) = (a - x)(a + x)$

$u \quad v$

(a) $f'(x) = u'v + uv' = -1(a+x) + (a-x) \cdot 1$

$= -a - x + a - x = -2x$

(b) $f(x) = (a-x)(a+x) = a^2 - x^2$ $\qquad (a-b)(a+b) = a^2 - b^2$

$f'(x) = -2x$

Aufgabe d)

- $\frac{d}{dx} \arccos(x) = \frac{-1}{\sqrt{1-x^2}}$

Regel: $g(x) = f^{-1}(x) \rightarrow g'(x) = \frac{1}{f'(g(x))}$ mit $f'(g(x)) \neq 0$

mit $f(x) = \cos(x) \rightarrow f'(x) = -\sin(x)$

wir haben: $g'(x) = \frac{1}{-\sin(g(x))} = \frac{1}{-\sin(\arccos(x))}$ $\quad \sin(\arccos(x)) = \sqrt{1-x^2}$

$= \frac{1}{-\sqrt{1-x^2}}$

$= \frac{-1}{\sqrt{1-x^2}} \rightarrow < 0$ (immer > 0) $\Rightarrow$ immer < 0

- $\frac{d}{dx} \arcsin(x) = ?$

$g(x) = \arcsin(x)$ $\quad g(x) = f^{-1}(x)$ und $g'(x) = \frac{1}{f'(g(x))}$ mit $f'(g(x)) \neq 0$

mit $f(x) = \sin(x)$
$f'(x) = \cos(x)$

wir haben: $g'(x) = \frac{1}{\cos(g(x))} = \frac{1}{\cos(\arcsin(x))}$ $\quad$ und $\cos(\arcsin(x)) = \sqrt{1-x^2}$

$= \frac{1}{\sqrt{1-x^2}}$

- $\boxed{\frac{d}{dx} \arcsin(x) = \frac{1}{\sqrt{1-x^2}}}$

Aufgabe 05

$$f(x) = e^{-x}\sin(x)$$

- $f'(x) = -e^{-x}\sin(x) + e^{-x}\cos(x)$
- $f''(x) = e^{-x}\sin(x) + (-e^{-x})\cos(x) + (-e^{-x})\cos(x) + e^{-x}\cdot(-\sin(x))$
  $= e^{-x}\left(\sin(x) - \cos(x) - \cos(x) - \sin(x)\right)$
  $= -2\cos(x)\,e^{-x}$
- $f'''(x) = 2\sin(x)\,e^{-x} + (-2\cos(x))\cdot -e^{-x}$
  $= e^{-x}\left(2\sin(x) + 2\cos(x)\right)$
  $= 2e^{-x}\left(\sin(x) + \cos(x)\right)$

Aufgabe 06

$$f(x) = a_0 + a_1 x + a_2 x^2$$

- $f(0) = -3 \iff a_0 + a_1 \cdot 0 + a_2 \cdot 0^2 = -3$
  $\iff a_0 = -3$

- $f'(x) = a_1 + 2a_2 x$
  $f'(0) = 5 \iff a_1 + 2a_2 \cdot 0 = 5$
  $\iff a_1 = 5$

- $f''(x) = 2a_2$
  $f''(x) = 4 \iff 2a_2 = 4 \iff a_2 = \frac{4}{2}$
  $\iff a_2 = 2$

$$\boxed{f(x) = -3 + 5x + 2x^2}$$

Aufgabe 7:

Wir haben: $f: \mathbb{R} \to \mathbb{R}, f(x) = \frac{x}{1+x^2}$

$$f'(x) = \frac{1 \cdot (1+x^2) - x \cdot 2x}{(1+x^2)^2} = \frac{1+x^2-2x^2}{(1+x^2)^2} = \frac{1-x^2}{(1+x^2)^2}$$

$$1 - x^2 = 0$$
$$1 = x^2$$

$x_1 = 1 \qquad x_2 = -1$ Kandidaten für

Also es gibt 2 Extremwerte: ein in $x = 1$ und ein andere wenn $x = -1$

$$f''(x) = \frac{(1+x^2)^4(-2x) - (1-x^2)2(1+x^2)\cdot 2x}{(1+x^2)^4}$$

$$= \frac{(1+x^2)\cdot(-2x) - (1-x^2)\cdot 4x}{(1+x^2)^3}$$

$$= \frac{2x^3 - 6x}{(1+x^2)^3}$$

$f''(x_1 = 1) < 0$ Also in $x=1$ es ist ein Maximum

$f''(x_2 = -1) > 0$ Also in $x=-1$ es ist ein Minimum

$2x^3 - 6x = 0 \quad \Leftarrow \quad f''(x) = 0$

$x(x^2 - 3) = 0$

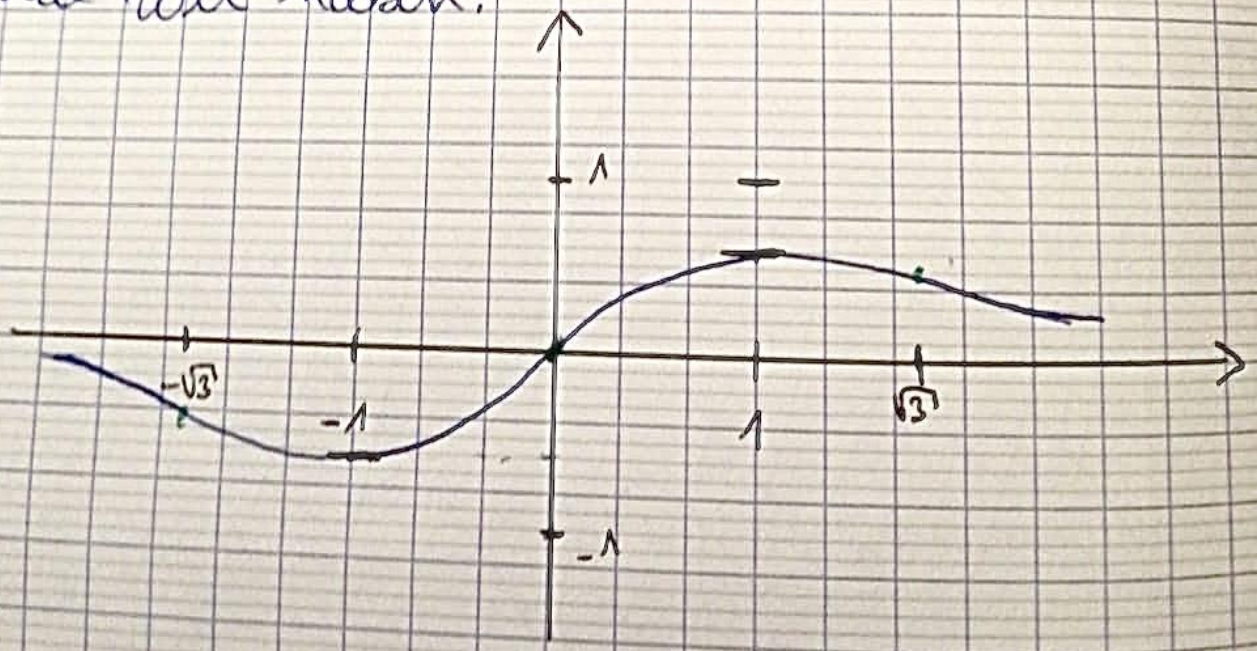
~~Es gibt eine Lösung~~

$x_0 = 0$

$x_1 = \sqrt{3}$

und $x_2 = -\sqrt{3}$ } Kandidaten für Wendepunkte $f(1) = \frac{1}{2}$ $f(-1) = -\frac{1}{2}$

Also wir haben:



Aufgabe 08

$$f(x) = u(x)^{v(x)} = \left(e^{\ln(u(x))}\right)^{v(x)}$$

$$= e^{\ln(u(x)) \cdot v(x)}$$

$$f'(x) = e^{\ln(u(x)) \cdot v(x)} \cdot \left(\ln(u(x)) \cdot v(x)\right)'$$

$$= e^{\ln(u(x)) \cdot v(x)} \cdot \left(\frac{u'(x)}{u(x)} \cdot v(x) + \ln(u(x)) \cdot v'(x)\right)$$

$$= \boxed{u(x)^{v(x)} \cdot \left(\frac{u'(x)}{u(x)} v(x) + \ln(u(x)) \cdot v'(x)\right)}$$

- $f(x) = x^x \quad \rightarrow u(x) = x \quad \rightarrow v(x) = x$

$$f'(x) = x^x \left[\ln(x) \cdot 1 + \frac{1}{x} \cdot x\right] = x^x \left[\ln(x) + 1\right]$$

- $f(x) = x^{-\ln(x)} \quad \rightarrow u(x) = x \quad \rightarrow v(x) = -\ln(x)$

$$f'(x) = x^{-\ln(x)} \left[\frac{1}{x} \cdot -\ln(x) + \ln(x) \cdot -\frac{1}{x}\right]$$

$$= x^{-\ln(x)} \left[-\frac{2\ln(x)}{x}\right]$$

$$= -\frac{2}{x} \ln(x)\, x^{-\ln(x)}$$

Aufgabe 09

$$G(p) = G_0(1+p)^n$$

(a) Wir suchen die Tangente $G_T(p) = mp + b$ an den Punkt $p=0$

$$G(0) = G_0(1+0)^n = G_0$$

und

$$G(0) = G_T(0) \iff G_0 = b$$

$$G'(p) = n G_0 (1+p)^{n-1} \quad \text{und} \quad G_T'(p) = m$$

$$G'(0) = n G_0$$

Also

$$G'(0) = G_T'(0) \iff n G_0 = m$$

$$\boxed{G_T(p) = n G_0 p + G_0 = G_0(np+1)}$$

(b) $G_0 = 1000\,€ \quad p = 0{,}02 \quad \text{und} \quad n = 5$

Nach 5 Jahren ist das Guthaben wert:

mit der exakten Funktion: $G(0{,}02) = 1000(1+0{,}02)^5 = 1104\,€$

mit der Tangente: $G_T(0{,}02) = 1000(5 \cdot 0{,}02 + 1) = 1100\,€$

$\Big\}$ $G(0{,}02) > G_T(0{,}02)$

Aufgabe 10:

a) $f(x) = 1 - e^{-x}(1+x)$

$f'(x) = -\left(-e^{-x}(1+x) + e^{-x}\right)$

$= e^{-x}(1+x) - e^{-x}$

$= x e^{-x}$

$f'(x) = 0 \quad$ für $x = 0$

$f''(x) = -e^{-x} + x e^{-x} = e^{-x}(x-1)$

$f''(0) = e^{-1} > 0 \quad$ also in $x = 0$ wir haben ein Wendepunkt (Minimum)

b) $t(x) = mx + b$ Wendetangente ist wenn $x = 1$ weil $f''(1) = 0$

$f'(1) = m = 1e^{-1}$

$f(1) = t(1) = m + b = 1 - 2e^{-1}$

$e^{-1} + b = 1 - 2e^{-1}$

$b = 1 - 3e^{-1}$

Also wir haben: $t(x) = \frac{x}{e} + 1 - \frac{3}{e}$

c) $\lim_{x \to +\infty} 1 - e^{-x}(1+x) = \lim_{x \to +\infty} 1 + \lim_{x \to +\infty} -e^{-x}(1+x) = 1 + 0 = 1$

Also $c = 1$

d) $t(x) = c = 1$

$\frac{x}{e} + 1 - \frac{3}{e} = 1$

$\frac{x}{e} - \frac{3}{e} = 0$

$\frac{x-3}{e} = 0$

$x = 3$

Aufgabe 11

$$O = 2\pi r^2 + 2\pi r h \qquad V = \pi r^2 h = 500\,ml$$

$$\Leftrightarrow \boxed{h = \frac{500\,ml}{\pi r^2}}$$

$$O(r) = 2\pi r^2 + 2\pi r h = 2\pi r^2 + 2\pi r \cdot \frac{500\,ml}{\pi r^2}$$

$$= 2\pi r^2 + \frac{1000\,ml}{r} \quad \} 2.V$$

$$O'(r) = 4\pi r - \frac{1000\,ml}{r^2}$$

Suche nach dem Minimum: $O'(r) = 0$

$$\Leftrightarrow 4\pi r - \frac{1000\,ml}{r^2} = 0$$

$$\Leftrightarrow 4\pi r = \frac{1000\,ml}{r^2}$$

$$\Leftrightarrow r^3 = \frac{1000\,ml}{4\pi}$$

$$\hookrightarrow r = \sqrt[3]{\frac{1000\,ml}{4\pi}}$$

und $1\,l = 1 \cdot 10^{-3}\,m^3$

$1000\,ml = 1 \cdot 10^{-3}\,m^3$

Also $r = \sqrt[3]{\frac{1 \cdot 10^{-3}\,m^3}{4\pi}} = 0{,}043\,m$

$$O''(r) = 4\pi + \frac{2 \cdot 1000\,ml}{r^3} = 4\pi + \frac{2000\,ml}{r^3}$$

$$O''(0{,}43) = 4\pi + \frac{2000\,ml}{(0{,}043)^3} > 0 \rightarrow \text{minimum}$$

und $h = \frac{0{,}5 \cdot 10^{-3}}{\pi \cdot (0{,}043)^2}$

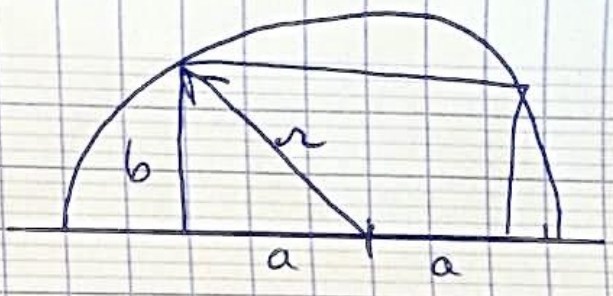
$= 0{,}086\,m$

$500\,ml = 0{,}5\,l$

$0{,}5\,l = 0{,}5 \cdot 10^{-3}\,m^3$

$\boxed{r = 4{,}3\,cm}$ $\boxed{h = 8{,}6\,cm}$

Aufgabe 12:



Mit Pythagoras wir haben:

$$a^2 + b^2 = r^2 \quad \Rightarrow \quad b = \sqrt{r^2 - a^2}$$

$$A(a) = 2a \cdot b = 2a \cdot \sqrt{r^2 - a^2}$$

$$A'(a) = 2\sqrt{r^2 - a^2} + 2a \cdot \frac{-2a}{2\sqrt{r^2 - a^2}}$$

$$A'(a) = 2\sqrt{r^2 - a^2} - \frac{2a^2}{\sqrt{r^2 - a^2}}$$

$$A'(a) = \frac{2(r^2 - a^2) - 2a^2}{\sqrt{r^2 - a^2}} = \frac{2r^2 - 4a^2}{\sqrt{r^2 - a^2}}$$

$$2r^2 - 4a^2 = 0$$

$$4a^2 = 2r^2$$

$$a^2 = \frac{2}{4} r^2$$

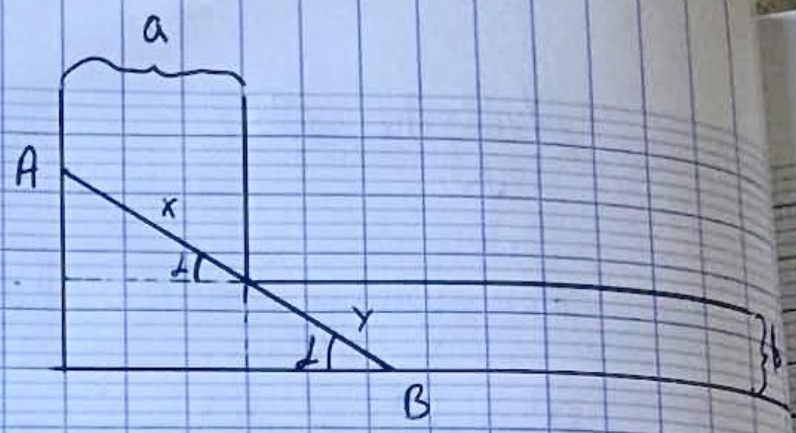
$$a^2 = \frac{1}{2} r^2$$

$$\boxed{a = \pm \frac{r}{\sqrt{2}}}$$

und $b = \sqrt{r^2 - \frac{1}{2}r^2} = \sqrt{\frac{1}{2}r^2} = \frac{r}{\sqrt{2}}$

$$\boxed{b = \frac{r}{\sqrt{2}}}$$

Aufgabe 13



Wir haben:

$\cos\alpha = \frac{a}{x}$ und $\sin\alpha = \frac{b}{y}$

Also: $x = \frac{a}{\cos\alpha}$ und $y = \frac{b}{\sin\alpha}$

$\Leftrightarrow f(\alpha) = \frac{a}{\cos\alpha} + \frac{b}{\sin\alpha}$

$f'(\alpha) = \frac{a\sin\alpha}{\cos^2\alpha} - \frac{b\cos\alpha}{\sin^2\alpha}$ Wir suchen für $f'(\alpha) = 0$

$a\sin^3\alpha = b\cos^3\alpha$

$\tan^3\alpha = \frac{b}{a}$

Also: $\alpha = \sqrt[3]{\arctan\left(\frac{b}{a}\right)}$

$\alpha \approx 3°$

$f''(\alpha) = +a \times 3\cos\alpha\sin^2\alpha + b \times 3\sin\alpha\cos^2\alpha > 0$

Also es ist ein minimum ✓

Aufgabe 14:

a) $\lim_{x \to 0} \frac{\sin(x)}{x} = \lim_{x \to 0} \frac{\cos(x)}{1} = \cos(0) = 1$

b) $\lim_{x \to 0} \frac{1}{x^2}\left(1 - \frac{1}{\cos x}\right) = \lim_{x \to 0} \frac{\frac{1}{\cos^2(x)} \sin x}{2x}$

$= \lim_{x \to 0} \frac{\frac{\sin(x)}{\cos^2(x)}}{2x}$

$= \lim_{x \to 0} \frac{\cos(x)}{2\cos^2(x) - 2x \cdot 2\cos(x)\sin(x)}$

$= \lim_{x \to 0} \frac{1}{2\cos(x) - 2x \cdot 2\sin(x)} = \frac{1}{2}$

c) $\lim_{x \to 0} \frac{\ln(2x+1)}{x} = \lim_{x \to 0} \frac{\frac{2x+1}{2}}{1}$

$= \frac{1}{2}$

d) $\lim_{x \to 0} \frac{\operatorname{arctan} x}{x} = \lim_{x \to 0} \frac{\frac{1}{1+x^2}}{1}$

$= 1$

e) $\lim_{x \to \infty} \frac{x^3}{e^x} = \lim_{x \to +\infty} \frac{3x^2}{e^x} = \lim_{x \to \infty} \frac{6x}{e^x} = \lim_{x \to +\infty} \frac{6}{e^x} = 0$